

Experimental Methods in Solid State Physics

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1 Crystal Growth

The quality of a solid-state experiment is inseparably linked to the quality of the crystal under investigation. Whether one aims to determine a lattice structure from diffraction data, measure charge carrier mobility in a transport setup, or resolve vibrational modes by spectroscopy, all of these rely on materials with high purity and low defect density. Poorly grown samples broaden diffraction peaks, obscure electronic states, and diminish transport properties. In many research programs, sample preparation is the true bottleneck: *no sample, no science*.

Impurities and defects are unavoidable in real crystals, but their concentration and distribution depend strongly on the growth route. **Impurities** (foreign atoms) may enter during synthesis from the growth chamber, the growth atmosphere, chemical transport agents, or incomplete purification of the source material. **Defects** (such as vacancies, dislocations, grain boundaries) are generated when the lattice fails to propagate perfectly during solidification, or when thermodynamic driving forces at elevated temperature favor a finite defect population. Growth parameters—temperature gradients, pulling and translation speeds, atmosphere, pressure, and cooling or annealing protocols—are chosen to minimize these imperfections or to drive them out after growth.

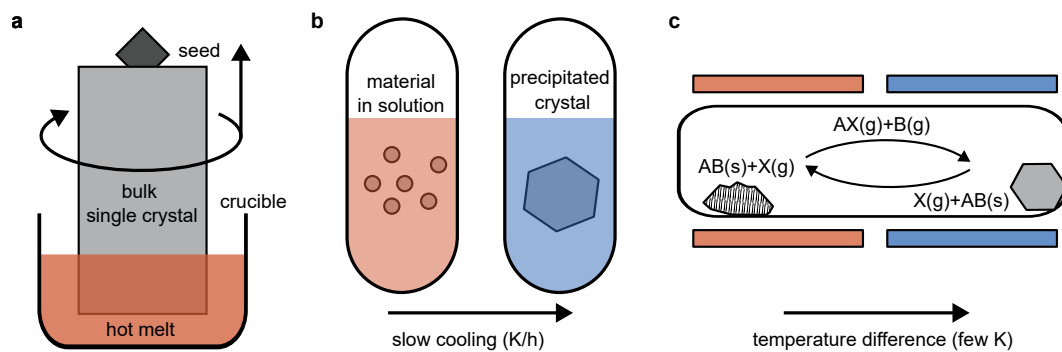


Figure 1: Examples of common crystal growth techniques. **a** Czochralski method. **b** Flux growth. **c** Chemical vapor transport.

Bulk crystal growth

Czochralski (CZ) method. A charge of the material is melted in a crucible, and a seed crystal is dipped into the melt and slowly pulled upward while rotating (Fig. 1a). Solidification proceeds at the solid–liquid interface, replicating the crystallographic order of the seed. The crystal diameter is controlled by a balance of heat flow and pulling rate; rotation helps homogenize the melt. CZ excels at producing large single crystals (e.g. silicon, germanium) and underpins wafer production in microelectronics. Technical challenges include thermal stresses that can nucleate dislocations, and contamination from the crucible (e.g. oxygen uptake from quartz in Si). Post-growth anneals and carefully tuned thermal gradients are employed to reduce defect densities and to control oxygen precipitation in downstream processing.

Equilibrium Vacancy Concentration

Even in an apparently perfect crystal, some lattice sites are empty. These *vacancies* are unavoidable because, at finite temperature, the crystal can lower its free energy by exchanging a small amount of internal energy for a large increase in entropy.

Energetic cost. Creating a single vacancy costs the *formation energy* E_f^v , which includes breaking the surrounding bonds and relaxing the lattice. Creating n vacancies raises the total internal energy by approximately

$$U(n) = n E_f^v.$$

Counting possibilities and entropy. The key point is that there are many ways to choose which n sites are vacant in a crystal with N atomic sites. The number of possible arrangements is

$$\Omega = \binom{N}{n} = \frac{N!}{n!(N-n)!}.$$

Therefore, a crystal with a small number of vacancies has a much higher entropy than a perfect crystal. The *configurational entropy* is defined as

$$S = k_B \ln \Omega,$$

where k_B is Boltzmann's constant. When N is very large (of order Avogadro's number) and $n \ll N$, we can use Stirling's approximation, $\ln N! \approx N \ln N - N$, to simplify:

$$S \approx k_B [N \ln N - n \ln n - (N - n) \ln (N - n)] \approx k_B n \left[\ln \left(\frac{N}{n} \right) + 1 \right].$$

This means that the entropy increases roughly like $k_B n \ln(N/n)$ —so adding a few vacancies greatly increases the number of accessible configurations.

Minimizing the total free energy. The relevant thermodynamic potential at fixed temperature is the Helmholtz free energy,

$$F(n) = U - TS \approx n E_f^v - k_B T n \left[\ln \left(\frac{N}{n} \right) + 1 \right].$$

Setting $\partial F / \partial n = 0$ gives

$$E_f^v - k_B T \ln \left(\frac{N}{n} \right) = 0, \quad \Rightarrow \quad \frac{n}{N} \equiv c_v = \exp \left(-\frac{E_f^v}{k_B T} \right).$$

In this simplified model, the equilibrium *vacancy fraction* is determined by a simple Boltzmann factor: sites with an energy cost E_f^v are occupied with a probability proportional to $e^{-E_f^v/k_B T}$.

Typical numbers. For $E_f^v = 1$ eV, we find

$$c_v(T=300 \text{ K}) \approx 10^{-17} \quad \text{and} \quad c_v(T=1200 \text{ K}) \approx 10^{-6}$$

Physical consequences. During crystal growth, the temperature is high and c_v correspondingly large. Upon cooling, the equilibrium defect density drops. If cooling is slow, defects have time to recombine or diffuse to sinks (dislocations, surfaces). Rapid quenching, in contrast, *freezes in* a nonequilibrium defect density. Controlling temperature profiles and annealing times is therefore one of the most powerful tools in obtaining crystals with low defect concentration.

Bridgman–Stockbarger method. Here, the material is sealed in an ampoule and translated slowly through a well-defined temperature gradient so that the charge melts and then resolidifies from one end to the other. Ideally a single seed determines the crystallographic orientation and the solidification front remains planar. The method is mechanically simpler than CZ and is widely used for compound semiconductors, e.g. GaAs, CdTe. Because the melt is confined, interactions with container walls can introduce impurities. Segregation during solidification may yield compositional inhomogeneities if the translation speed and gradient are not optimized. Grain formation is suppressed by maintaining a stable interface and modest growth rates; too steep a gradient or too fast a pull promotes polycrystallinity and dislocation generation.

Flux growth. Flux growth exploits a high-temperature solvent (the “flux”) in which the target compound has appreciable solubility. Upon slow cooling, the solution becomes supersaturated and the compound crystallizes out, ideally on a seed or as free-standing single crystals (Fig. 1b). Because dissolution and precipitation occur well below the melting point of the compound, flux growth is especially valuable for materials that decompose or melt incongruently. Choice of flux is central: a *self-flux* (one component of the target stoichiometry) minimizes chemical contamination, whereas foreign-metal fluxes (e.g. Sn, Bi) can widen the accessible solvent window but risk incorporation or surface wetting. In practice, a charge composed of starting elements (or pre-reacted powders) and flux is sealed in a crucible, heated to homogenize, then cooled at very slow cooling rates (typically 0.1 – 0.5 K/h) through the supersaturation regime.

Chemical Vapor Transport (CVT). Chemical vapor transport grows crystals by *reversible* gas-phase reactions that shuttle material from a hot source zone to a cooler sink zone within a sealed ampoule (Fig. 1c). A small amount of transport agent X (e.g. halogens or halides) reacts with the source compound, denoted AB, at temperature T_{hot} to form volatile species AX(g) and B(g) into the gas phase; these diffuse to the cold zone at T_{cold} , where the equilibrium shifts and the target compound AB re-precipitates in solid form, releasing the transport agent X into the gas phase. The net flux is set by the gradient in chemical potential between zones and the gas-phase diffusivity. CVT is particularly effective for layered chalcogenides and many transition-metal compounds, where direct melt growth is impractical or yields polycrystals. Both flux growth and chemical vapor transport achieve equilibrium crystal growth far below the melting point, which improves chemical homogeneity at the expense of growth speed.

Thin-film growth

Many materials and device structures used in modern technology cannot be grown as large bulk crystals but instead require atomically precise thin films or heterostructures. These are obtained by epitaxial growth techniques, in which atoms or molecules condense on a single-crystalline substrate and arrange themselves in registry with its lattice. The term *epitaxy* (from Greek *epi*, “upon”, and *taxis*, “arrangement”) emphasizes that the crystallographic orientation of the film is determined by the substrate. Depending on the source of the depositing species, epitaxy can be realized in ultra-high vacuum (*molecular beam epitaxy*, MBE) or in a chemically reactive gas environment (*metal–organic chemical vapor deposition*, MOCVD). Thin-film growth unites thermodynamics and kinetics: surface energies decide whether atoms wet or cluster, while lattice mismatch and strain determine how many coherent layers can be sustained. Techniques such as MBE and MOCVD translate these principles into atomic-scale control, enabling modern semiconductor technology and the exploration of quantum phenomena in engineered heterostructures.

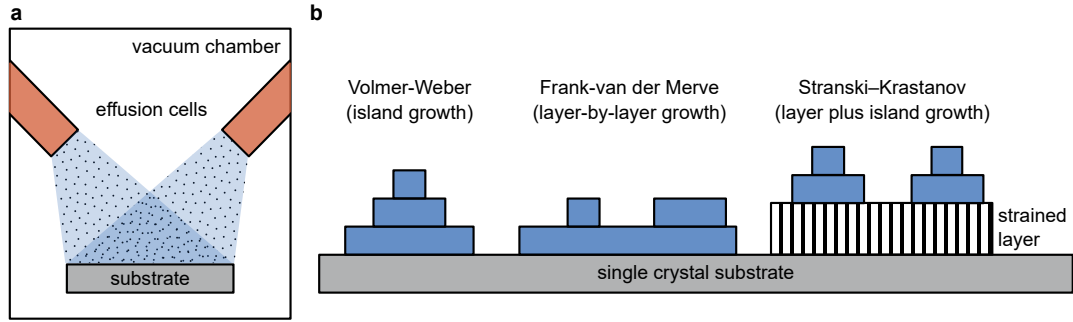


Figure 2: **a** Schematic illustration of thin-film growth via molecular beam epitaxy. **b** Comparison of the three basic thin-film growth modes.

Molecular Beam Epitaxy (MBE). In MBE, beams of atoms or molecules are evaporated from effusion cells under ultra-high-vacuum conditions ($p \lesssim 10^{-8}$ mbar) and directed toward a heated substrate (Fig. 2a). Because the mean free path is much larger than the chamber dimensions, the particles travel ballistically, allowing excellent control over flux and composition. Layer thickness is determined by the arrival rate and substrate temperature, and can be monitored in real time by electron diffraction methods.

The low background pressure minimizes contamination, leading to extremely pure interfaces and sharp heterojunctions. However, maintaining lattice matching between film and substrate is essential to prevent misfit dislocations. For small mismatches, strain can be elastically accommodated over several monolayers; beyond a critical thickness, plastic relaxation occurs and dislocations nucleate to relieve the strain energy. Precise flux control and growth interruptions at submonolayer level enable the fabrication of quantum wells, superlattices, and two-dimensional electron gases central to modern semiconductor physics.

Metal–Organic Chemical Vapor Deposition (MOCVD). MOCVD achieves epitaxial growth through chemical reactions of volatile precursor molecules at a heated substrate in a controlled atmosphere (typically 10^{-1} – 10^2 mbar). Organometallic compounds such as trimethylgallium or diethylzinc deliver the metal component, while hydrides such as AsH_3 or PH_3 supply nonmetals. At the substrate temperature (600°C to 1000°C), these precursors decompose, releasing the constituent atoms, which incorporate into the growing film while gaseous by-products are pumped away.

Compared to MBE, MOCVD operates at higher pressures and is less demanding in vacuum technology, making it suitable for industrial-scale wafer growth. It remains the standard for compound semiconductors such as GaAs, InP, and GaN, used in LEDs and high-frequency electronics. The principal challenges lie in controlling gas-phase reactions, precursor depletion, and uniformity across large substrates. Because chemical reactions can generate carbon or oxygen residues, purification of precursors and precise temperature control are critical for achieving low defect densities.

Growth modes. Commonly, thin-film growth can be classified into three distinct mechanisms, which are island growth, layer-by-layer growth, and layer-plus-island growth (Fig. 2b). Whether a film grows layer-by-layer or as isolated islands is determined by the balance of surface and interface energies. Let γ_s , γ_f , and γ_i denote the surface energies of substrate, film, and the interface,

respectively. If

$$\gamma_s > \gamma_f + \gamma_i,$$

atoms of the film wet the substrate completely, resulting in **Frank–van der Merwe** (layer-by-layer) growth. If

$$\gamma_s < \gamma_f + \gamma_i,$$

film atoms bond more strongly to each other than to the substrate, producing **Volmer–Weber** (island) growth. Intermediate cases, where the first few monolayers wet the surface but strain accumulates, lead to **Stranski–Krastanov** growth: initial layer-by-layer deposition followed by island formation once the strain energy exceeds the wetting energy gain. Control of these energies—by adjusting substrate temperature, background gas, and surface reconstruction—is the key to tailoring film morphology in epitaxy.

Strain and Critical Thickness

When a crystalline film is grown on a substrate with a slightly different lattice constant, the atomic planes of the film must stretch or compress to match those of the substrate. The relative mismatch,

$$\varepsilon = \frac{a_f - a_s}{a_s},$$

defines the in-plane strain. For very thin films, this distortion can be accommodated elastically: the film simply adjusts its lattice spacing without introducing defects. Such a film is called *pseudomorphic* or *coherently strained*.

As the film becomes thicker, the elastic energy stored in the lattice increases proportionally to the film thickness, because each additional layer adds more strained material. Eventually, it becomes energetically favorable for the film to relax by forming small lattice imperfections at the interface, such as dislocations or stacking faults. The film thickness at which this happens is called the *critical thickness* h_c .

A detailed elastic analysis (beyond the scope of this tutorial) shows that the critical thickness decreases approximately inversely with the lattice mismatch:

$$h_c \propto \frac{1}{|\varepsilon|}.$$

This means that even a small mismatch drastically limits the thickness of a defect-free epitaxial layer.

To get a feeling for typical numbers, consider GaAs grown on an AlAs substrate, which has a very small lattice mismatch of only $|\varepsilon| \approx 0.001$. In this case, coherent layers up to several hundred nanometers thick can be grown without strain relaxation. In contrast, for GaN grown on sapphire, the mismatch is about $|\varepsilon| \approx 0.04$ (4%), and the corresponding critical thickness is only a few nanometers. Beyond that, misfit dislocations form to relieve the strain, leading to mosaicity and reduced carrier mobility.

In practice, modern epitaxy exploits this interplay between strain and relaxation. For example, thin strained layers can be used deliberately to tune band gaps and enhance carrier mobility (“strain engineering”), while thicker films require graded buffer layers or van der Waals substrates to avoid defect formation.

Question 1 *Defect density*

At $T = 1200\text{ K}$, two point defects have formation energies $E_f = 2\text{ eV}$ and $E_f = 5\text{ eV}$. Assuming $c \propto \exp[-E_f/(k_B T)]$ with $k_B = 8.6 \times 10^{-5}\text{ eV/K}$, what is the approximate ratio of their equilibrium concentrations c_2/c_5 ?

- (a) 1
- (b) $e^{-(5-2)/(8.6 \times 10^{-5} \times 1200)} \approx 10^{-13}$
- (c) $e^{(5-2)/(8.6 \times 10^{-5} \times 1200)} \approx 10^{13}$
- (d) $1/(3 \times 1200) \approx 3 \times 10^{-4}$

Question 2 *Bulk crystal growth*

During Czochralski growth of silicon, the most important source of oxygen contamination is:

- (a) evaporation from the melt surface.
- (b) residual gas in the chamber.
- (c) the rotating seed crystal.
- (d) reaction and dissolution of the quartz crucible at high temperature.

Question 3 *Bulk crystal growth*

Which statement regarding *flux growth* vs. *chemical vapor transport* (CVT) is correct?

- (a) Both operate near the melting point and typically yield the largest crystals.
- (b) Flux growth relies on supersaturation in a liquid solvent (diffusion in melt); CVT uses a gas-phase transport agent and a hot-cold equilibrium to drive deposition.
- (c) CVT growth minimizes contamination better than flux growth because the halogens used as transport agents purify the source material.
- (d) CVT always produces lower defect density due to lower temperatures.

Question 4

Thin-film growth A GaAs film ($a_f = 5.65\text{ \AA}$) is grown on a GaP substrate ($a_s = 5.45\text{ \AA}$). Estimate the lattice mismatch ε and the qualitative implication for the critical thickness.

- (a) $|\varepsilon| \approx 0.2\%$; coherent thickness $\sim$ hundreds of nm.
- (b) $|\varepsilon| \approx 3.7\%$; coherent growth limited to only a few nm before misfit dislocations.
- (c) $|\varepsilon| \approx 8\%$; film remains pseudomorphic up to micrometer.s
- (d) $|\varepsilon| \approx 10^{-4}$; strain negligible.

2 Real-Space Probes I: Scanning tunneling and atomic force microscopy

Optical microscopy is limited in resolution by the wavelength of light: even under ideal conditions, features smaller than roughly $\lambda/2$ (a few hundred nanometers) cannot be resolved. Real space atomic details of surfaces remained inaccessible until the 1980s, when a new class of instruments—*scanning probe microscopes*—overcame this limit by exploiting interactions between a sharp tip and a surface at the atomic scale.

The first of these was the *scanning tunneling microscope* (STM), invented by G. Binnig and H. Rohrer in 1981. It provided the first direct images of individual atoms on a solid surface and earned its inventors the 1986 Nobel Prize in Physics. A few years later, the related *atomic force microscope* (AFM) extended the idea to non-conductive samples, making scanning probe techniques central to modern nanoscience.

Scanning Tunneling Microscopy (STM)

Principle of operation. The STM operates by bringing a sharp metallic tip extremely close to a conductive surface—so close that the outermost atoms of tip and sample are separated by only a few ångströms. When a bias voltage V is applied between them, electrons can *tunnel* quantum mechanically through the vacuum gap (Fig. 3a). The resulting tunneling current I depends sensitively on the tip-sample separation z : even a displacement of 1 Å can change the current by roughly one order of magnitude. This exponential dependence makes it possible to map the surface with atomic-scale resolution.

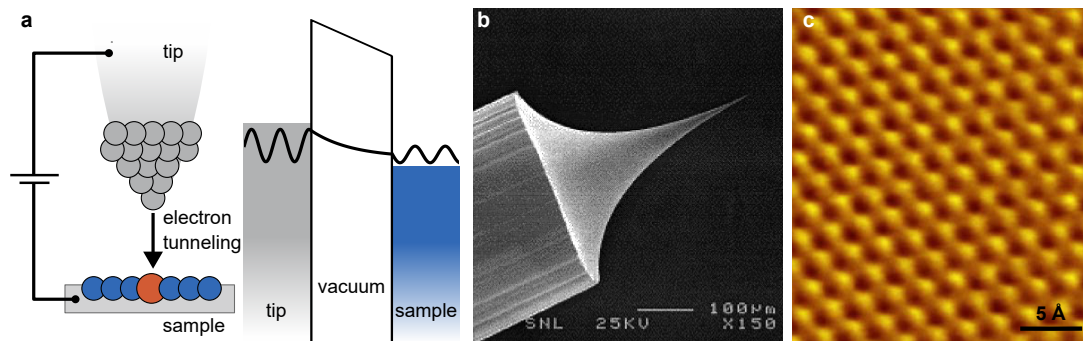


Figure 3: Scanning tunneling microscopy (STM). **a** Working principle of STM. **b** Electron microscope image of an electrochemically etched tungsten tip. Reprinted with permission from Meas. Sci. Technol. 10, 11–18 (1999). Copyright IOP 1999. **c** Constant current image of the surface of highly oriented pyrolytic graphite (HOPG) resolving the hexagonal periodic lattice structure. Reprinted with permission from Solid State Commun. 149, 1151–1156 (2009). Copyright Elsevier 2009.

Technical implementation. The STM tip is mounted on a piezoelectric scanner capable of sub-ångström motion in all three directions. The scanner raster-scans the tip across the surface while a feedback loop continuously adjusts its height to maintain a constant tunneling current. The resulting vertical displacement $z(x, y)$ forms the topographic map of the surface. This *constant-current mode* is the most common operating mode and provides stable, high-resolution images.

An alternative is the *constant-height mode*, in which the tip height remains fixed while variations in tunneling current are recorded. This mode is faster but riskier—any protrusion higher than the set distance can cause a tip crash. Constant-height operation is therefore restricted to atomically flat surfaces such as graphite or semiconductor terraces.

The quality of an STM image depends critically on the sharpness and cleanliness of the tip. In practice, tips are usually made from tungsten or platinum–iridium wires a few tenths of a millimeter thick. Tungsten tips are often prepared by *electrochemical etching*, in which a short wire is immersed in a sodium hydroxide solution and biased against a counter electrode. When the wire thins to a fine neck, it breaks spontaneously, leaving a sharp apex with a radius of curvature of only a few nanometers (Fig. 3b). The freshly etched tip is rinsed, sometimes annealed in vacuum, and occasionally further sharpened by field evaporation or gentle indentation into a clean metal surface.

Quantum Tunneling and STM Imaging Contrast

For an electron with energy E incident on a 1D potential barrier of height Φ and width z , quantum mechanics predicts a finite transmission probability

$$T \propto \exp(-2\kappa z), \quad \text{with} \quad \kappa = \frac{\sqrt{2m(\Phi - E)}}{\hbar}.$$

In the STM, this leads to a tunneling current approximately proportional to

$$I(V, z) \propto V \rho_s(E_F) e^{-2\kappa z},$$

where $\rho_s(E_F)$ is the *local density of electronic states* (LDOS) of the sample at the Fermi energy. The STM signal therefore depends on both the geometric distance and the electronic structure at the surface.

When a bias voltage V is applied, electrons tunnel from occupied states of one electrode to unoccupied states of the other. The current can be expressed as

$$I(V) \propto \int_{E_F}^{E_F + eV} \rho_s(\mathbf{r}, E) T(E, z) dE,$$

which means that changing the bias allows one to probe occupied ($V < 0$) or empty ($V > 0$) electronic states. In a constant-current image, regions with higher LDOS at the chosen energy appear as apparent “protrusions” because the feedback system retracts the tip slightly to keep I constant. STM images therefore represent contours of constant integrated LDOS rather than purely geometric height maps.

Atomic Force Microscopy (AFM)

While the STM relies on a tunneling current and thus requires a conductive sample, the *atomic force microscope* (AFM) generalizes the concept of scanning with a sharp tip to measure mechanical rather than electronic interactions. In AFM, a sharp tip mounted at the end of a flexible cantilever is brought close to the sample surface. The forces acting between tip and surface—arising from van der Waals attraction, Pauli repulsion, and possibly electrostatic or capillary interactions—bend the cantilever by minute amounts. These deflections, typically on the order of picometers to nanometers, are measured with sub-ångström precision using an optical lever: a laser beam reflected from the

cantilever onto a position-sensitive, segmented photodiode records its angular motion.

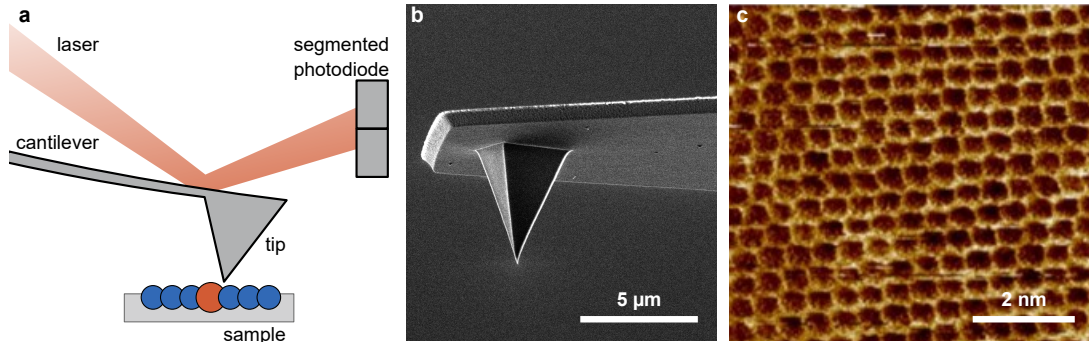


Figure 4: Atomic force microscopy (AFM). **a** Working principle of AFM. **b** Electron microscope image of a pyramid-shaped tip on a microcantilever fabricated from silicon. Adapted under the terms of Creative Commons Attribution-Share Alike 4.0 International from wikimedia.org. Copyright NanoscaleRanger 2019. **c** Atomically resolved topography of a Mica surface, an insulating, layered silicate mineral. Adapted with permission from Rev. Sci. Instrum. 81, 073704 (2010). Copyright AIP 2010.

Lennard–Jones Potential

The interaction between two atoms or surfaces can often be modeled by a Lennard–Jones potential

$$U(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right],$$

where ε is the binding energy and σ the equilibrium separation. The corresponding force is

$$F(r) = -\frac{dU}{dr} = 24\varepsilon \left[\frac{2\sigma^{12}}{r^{13}} - \frac{\sigma^6}{r^7} \right].$$

At large r , the attractive r^{-7} term dominates (van der Waals regime); at small r , the steep r^{-13} term causes repulsion due to overlapping electron orbitals. The transition between these regimes determines whether the AFM operates in attractive (non-contact) or repulsive (contact) mode.

Operating principle and force regimes. As the tip approaches the surface, the interaction passes smoothly from long-range attraction to short-range repulsion. The resulting *force–distance curve* governs the contrast in all AFM modes. In *contact mode*, the tip lightly touches the surface on the repulsive side of the curve, and topography is recorded by maintaining a constant deflection. In *non-contact mode*, the cantilever oscillates near its resonance frequency a few nanometers above the surface; attractive forces reduce its effective spring constant and cause a measurable frequency shift. *Tapping* or *intermittent-contact mode* lies between these two limits and is widely used because it minimizes lateral forces and sample damage.

Nanomechanics with AFM. Beyond imaging, the AFM provides a quantitative tool for exploring mechanical properties at the nanoscale. When the tip indents a surface, the local deformation obeys the same continuum concepts introduced in solid mechanics: stress, strain, and elastic moduli. In the simplest case of a spherical tip of radius R pressing into an elastic half-space, the contact radius a and indentation depth δ are related by the *Hertz model*,

$$F = \frac{4}{3}E^*\sqrt{R}\delta^{3/2}, \quad \text{with} \quad \frac{1}{E^*} = \frac{1 - \nu_t^2}{E_t} + \frac{1 - \nu_s^2}{E_s},$$

where $E_{t,s}$ and $\nu_{t,s}$ are the Young's moduli and Poisson ratios of tip and sample, respectively. Measuring the force–distance curve thus allows extraction of the effective elastic modulus E^* .

Typical AFM cantilevers have spring constants between 10^{-2} and 100 N/m. A deflection of only 1 nm therefore corresponds to forces from piconewtons to nanonewtons—comparable to the weight of a single bacterium or the force between two chemical bonds. Conversely, when a sharp tip contacts a surface over a few square nanometers, these small forces translate into enormous local pressures of several gigapascals, sufficient to deform or even manipulate individual atoms. This remarkable range makes the AFM not only a microscope but also a nanoscale mechanical tester that directly links macroscopic elasticity to atomic-scale interactions.

Driven Harmonic Oscillator and Frequency Shift

A cantilever of effective mass m and spring constant k obeys

$$m\ddot{z} + \gamma\dot{z} + kz = F_0 \cos(\omega t),$$

with resonance frequency $f_0 = \frac{1}{2\pi}\sqrt{k/m}$ and quality factor $Q = \frac{1}{\gamma}\sqrt{km}$. When the tip experiences a weak additional force gradient $\partial F/\partial z$, the effective spring constant becomes $k_{\text{eff}} = k - \partial F/\partial z$, leading to a frequency shift

$$\Delta f \approx -\frac{f_0}{2k} \frac{\partial F}{\partial z}.$$

Attractive forces ($\partial F/\partial z < 0$) decrease f_0 , while repulsive interactions increase it. In non-contact AFM, the feedback loop maintains a constant oscillation amplitude or frequency shift, allowing topographic mapping with minimal tip–sample contact and atomic-scale resolution.

Question 5 *Tunneling in STM*

In an STM, the tunneling current follows approximately $I \propto e^{-2\kappa z}$ with $\kappa \approx 10 \text{ nm}^{-1}$. By how much does the current change if the tip-sample distance increases by $1 \text{ \AA}$?

- (a) The current doubles.
- (b) The current decreases by about one order of magnitude.
- (c) The current remains constant.
- (d) The current drops to zero instantly.

Question 6 *Image contrast in STM*

In constant-current STM imaging, the feedback loop adjusts the tip height so that the tunneling current remains fixed. What does the recorded topographic signal actually represent?

- (a) The true geometric height of the outermost surface atoms.
- (b) The local electric potential at the surface.
- (c) Contours of constant integrated local density of electronic states (LDOS) at the chosen bias voltage.
- (d) The spatial variation of the work function of the material.

Question 7 *Forces in AFM*

The interaction between an AFM tip and a surface atom can be approximated by a Lennard-Jones potential

$$U(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right].$$

At which distance regime is the tip-sample force attractive, and how does this relate to AFM imaging modes?

- (a) For $r < \sigma$, the force is attractive; AFM operates in non-contact mode.
- (b) For $r > \sigma$, the van der Waals term dominates and the force is attractive; AFM operates in non-contact mode.
- (c) For $r > \sigma$, the force becomes repulsive; AFM operates in contact mode.
- (d) The force is always repulsive because surfaces cannot attract.

Question 8 *Comparing STM and AFM*

Which statement correctly compares STM and AFM?

- (a) STM measures mechanical forces, while AFM measures tunneling current.
- (b) STM requires a conductive sample and measures electron tunneling; AFM measures interatomic forces and can image insulating materials.
- (c) Both techniques require vacuum and cryogenic temperatures.
- (d) AFM generally achieves higher lateral resolution than STM on metal surfaces.

3 Real-Space Probes II: Transmission Electron Microscopy

After exploring scanning probe methods that map surfaces atom by atom, we now turn to a technique capable of imaging *within* solids. The transmission electron microscope (TEM) provides real-space information from thin crystalline specimens and can reach sub-ångström resolution, sufficient to resolve individual atomic columns deep in the bulk. Whereas STM and AFM are limited to the outermost atomic layers, TEM reveals the internal structure, defects, and interfaces of materials.

Principle of operation

The concept of using electrons instead of light for microscopy dates back to the early 1930s. Because electrons exhibit wave-particle duality, Ernst Ruska realized that their wavelength could be much shorter than that of visible light, and that electromagnetic lenses could be used to focus them just as glass lenses focus light. Together with Max Knoll, he built the first transmission electron microscope in 1931 and demonstrated a spatial resolution surpassing optical microscopes. For this pioneering work, Ruska was later awarded the Nobel Prize in Physics in 1986—the same year that Gerd Binnig and Heinrich Rohrer received the prize for inventing the STM. Both breakthroughs transformed our ability to “see” matter on the atomic scale, one through electron optics, the other through quantum tunneling.

In a TEM, a high-energy electron beam—typically 80 keV to 300 keV—is generated by a thermionic or field-emission gun and guided down an evacuated electron optics column. A series of electromagnetic condenser lenses shape and focus the beam onto the sample, typically a thinned slab of crystalline material. The sample must be sufficiently electron-transparent (50 nm to 100 nm) so that a significant fraction of the electrons can traverse it without complete absorption. After interaction with the sample, the transmitted and scattered electrons are refocused by the objective and projection lenses to form a magnified image or diffraction pattern. In former days the pattern was detected by a fluorescent screen, whereas nowadays modern semiconductor cameras with few-electron sensitivity are well established.

Interaction of electrons with matter. When electrons penetrate a crystalline solid, they scatter with both atomic nuclei and the surrounding electron cloud:

- **Elastic scattering:** the electron changes direction but not its kinetic energy. It arises from the Coulomb potential of the atomic cores and is responsible for diffraction and image contrast. The angular distribution of this scattering carries structural information about the arrangement of atoms. At high accelerating voltages, the elastic mean free path (typically 50 nm to 200 nm) is long enough for thin specimens to transmit a measurable fraction of the incident beam.
- **Inelastic scattering:** the electron transfers energy to the sample, exciting vibrations (phonons), collective excitations (e.g. plasmons), or ionizing electrons. This process leads to energy loss, heating, and beam damage, and limits the total dose a specimen can tolerate. Inelastic scattering also reduces image contrast by adding background intensity.

The requirement of thin samples therefore stems from the balance between these two processes: the crystal to be imaged must be thick enough to represent the bulk but thin enough that most electrons experience at most one significant scattering event before leaving the sample. This competition between scattering and absorption governs both image contrast and achievable thickness.

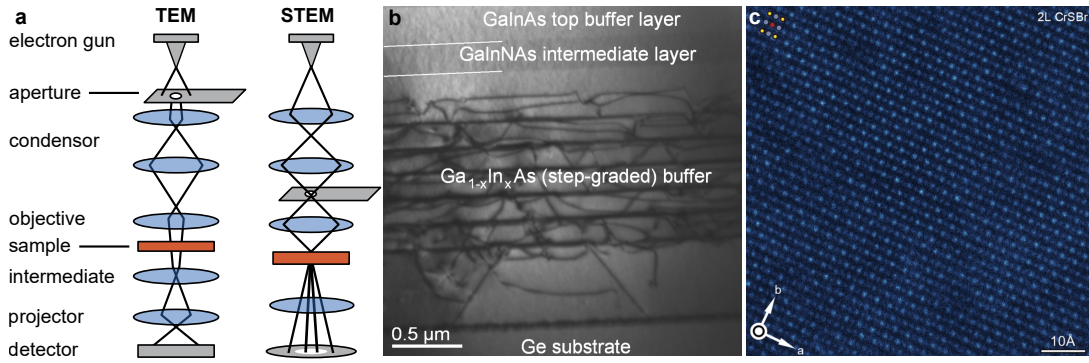


Figure 5: Transmission electron microscopy (TEM) and scanning transmission electron microscopy (STEM). **a** Working principle of TEM and STEM. **b** Bright-field image of an epitaxially grown semiconductor heterostructure revealing the formation of dislocations (dark contrast) due to strain relaxation. Reprinted from J. Phys.: Conf. Ser. 471 012008 (2013) under the terms of Creative Commons Attribution 3.0 license. Copyright The Authors 2013. **c** Atomically resolved lattice of a thin bilayer film of CrSBr, an antiferromagnetic van der Waals material. Reprinted from Phys. Rev. X 15, 021080 (2025) under the terms of Creative Commons Attribution 4.0 International license. Copyright The Authors 2025.

Electron Wavelength and Resolution Limit

The resolving power of a microscope depends on the wavelength of its probe. An electron accelerated through a potential V attains a relativistic momentum

$$p = \sqrt{2m_e eV \left(1 + \frac{eV}{2m_e c^2} \right)},$$

so that its de Broglie wavelength is

$$\lambda = \frac{h}{p}.$$

At 200 keV, $\lambda \approx 2.5 \times 10^{-3} \text{ nm}$ (0.0025 Å), about five orders of magnitude shorter than visible light. Even though lens aberrations prevent achieving this theoretical limit, practical spatial resolutions below 0.1 nm —corresponding to atomic spacing—are nowadays routinely achieved.

Electron optics and image formation. The principles of electron optics closely mirror those of geometric optics. Just as in an optical microscope, image formation in a TEM follows the principles of geometric optics. Magnetic lenses—coils of wire producing an axially symmetric magnetic field—act on moving electrons via the Lorentz force, bending their trajectories toward the optical axis much like a convex glass lens focuses light rays. A typical TEM employs several such lenses in sequence: the *condenser lens* controls the beam convergence and illumination area, the *objective lens* forms the first magnified image of the specimen, and the *intermediate* and *projector lenses* further enlarge this image onto the viewing screen or detector (Fig. 5a). The overall magnification is approximately the product $M = M_{\text{objective}} M_{\text{intermediate}} M_{\text{projector}}$. Since the individual magnifications are each

on the order of $100\times$, the total magnification often exceeds 10^5 . Modern TEMs therefore combine enormous geometric magnification with the intrinsically short electron wavelength to achieve atomic-scale resolution, revealing e.g. the formation of dislocations within semiconductor heterostructures (Fig. 5b).

Numerical aperture and collection angle. In analogy to optical microscopy, the resolution of a TEM also depends on the range of scattering angles collected by the objective lens, often expressed as a *numerical aperture* (NA). Formally, $NA = n \sin \alpha$, where α is the semi-angle of the objective lens aperture and n the refractive index of the medium (unity in vacuum). Because the electron wavelength is extremely small, even a modest angular aperture of a few milliradians corresponds to very high resolving power. However, increasing α also enhances lens aberrations and reduces contrast by admitting more highly scattered electrons. In practice, the objective aperture therefore acts as a spatial filter: by selecting a smaller collection angle, one emphasizes mass-thickness and diffraction contrast, whereas a larger aperture favors high-resolution, phase-contrast imaging. Thus, while the NA in TEM formally plays the same role as in light optics, its optimization reflects a compromise between resolution, contrast, and aberration control rather than simple diffraction limits.

Aberration correction In any lens system, spherical and chromatic aberrations distort the focus of off-axis rays, limiting resolution. In early TEMs, the spherical aberration coefficient C_s of the objective lens defined a point-spread function with radius proportional to $(C_s \lambda^3)^{1/4}$. The advent of multipole correctors—combinations of hexapole or octupole lenses that introduce compensating fields—has revolutionized the field. The corrector effectively flattens the phase front of high-angle rays, restoring them to the same focal plane as paraxial electrons. Corrected TEMs routinely achieve sub-ångström probes with minimal phase distortion, allowing direct imaging of atomic columns (Fig. 5c).

Z-Contrast and Incoherent Scattering

For an electron scattered elastically by a Coulomb potential, the Rutherford differential cross-section is

$$\frac{d\sigma}{d\Omega} \propto \frac{Z^2}{E^2 \sin^4(\theta/2)},$$

so the integrated intensity at large scattering angles grows roughly with Z^2 . In HAADF-STEM, this incoherent scattering dominates the detected signal, producing a direct, monotonic relationship between brightness and atomic number—unlike conventional phase-contrast TEM, where interference effects complicate interpretation.

HAADF-STEM imaging

Whereas conventional TEM relies on interference between transmitted and diffracted beams, scanning TEM (STEM) analyzes electrons scattered at defined angles, providing new types of image contrast. TEM forms images using the entire transmitted beam (bright-field) or selected diffracted beams (dark-field). In modern STEM instruments, the beam is instead focused into a fine probe and raster-scanned across the sample. Detectors placed at different scattering angles collect the transmitted electrons to form images with various contrast mechanisms.

A particularly powerful mode is the *high-angle annular dark-field* (HAADF) STEM, where electrons scattered to large angles (tens of milliradians) are recorded by an annular detector (Fig. 5a).

Because high-angle scattering arises from incoherent Rutherford-type interactions with the atomic nuclei, the resulting signal intensity scales approximately as $I \propto Z^{1.7-2}$, where Z is the atomic number. This Z -contrast makes heavy atoms appear bright and enables chemical contrast at atomic resolution.

Sample preparation

Achieving high-resolution imaging requires electron-transparent specimens. Bulk samples are thinned mechanically to a few tens of micrometers, then ion-milled or etched to perforation. For site-specific studies, a *focused ion beam* (FIB) microscope can lift out and thin cross-sectional lamellae directly from devices or heterostructures. Typical final thicknesses are 30 nm to 80 nm, striking a balance between minimizing multiple scattering and maintaining structural integrity. Surface contamination and beam damage are minimized by gentle low-energy cleaning immediately before insertion into the TEM.

Question 9 *Image contrast*

In a high-angle annular dark-field (HAADF) STEM image, which statement about image brightness is correct?

- (a) Boron ($Z = 5$) will appear brighter than carbon ($Z = 6$).
- (b) Carbon ($Z = 6$) will appear slightly brighter than boron ($Z = 5$).
- (c) Both elements appear with equal brightness because they are adjacent in the periodic table.
- (d) Image brightness does not depend on atomic number but only on surface roughness.

Question 10 *Sample preparation*

Why must TEM specimens typically be thinner than about 100 nm?

- (a) To prevent magnetic deflection of the electron beam.
- (b) Because the vacuum system cannot accommodate thicker samples.
- (c) To minimize multiple elastic and inelastic scattering and maintain interpretable, high-contrast transmission images.
- (d) To reduce charging effects due to secondary electron emission.

Question 11 *STEM*

In high-angle annular dark-field (HAADF) STEM imaging, the contrast is mainly determined by:

- (a) Interference between transmitted and diffracted beams.
- (b) Incoherent Rutherford-type scattering.
- (c) Bragg reflection from specific lattice planes.
- (d) Secondary electron emission from the specimen surface.